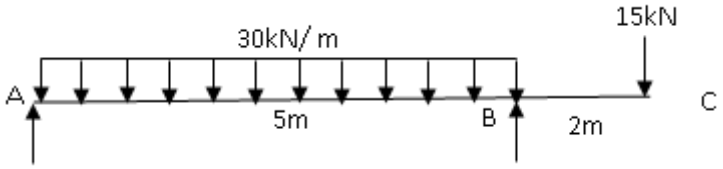
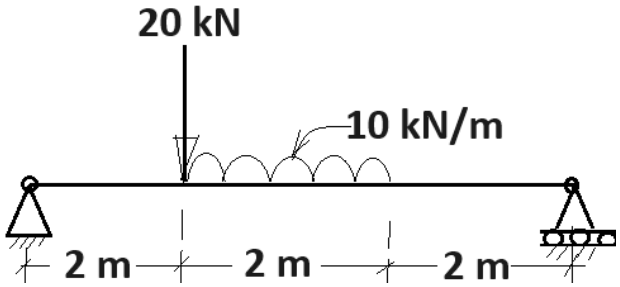
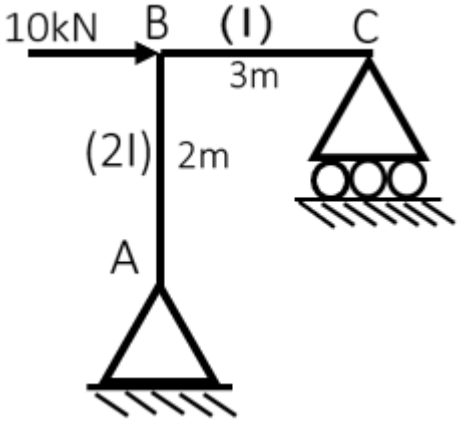
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(A constituent unit of MAHE, Manipal)

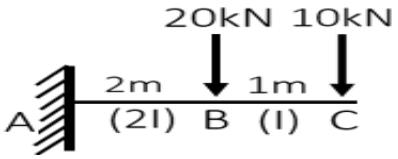
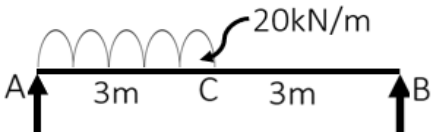
DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOV 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**CIVIL ENGINEERING**	**Course Type**	**C**
Course Name	**MECHANICS OF STRUCTURES**	**Course Code**	**2124**
Semester	**III**	**Max Marks**	**50**
Date	**22-11-2024**	**Duration**	**3 HRS**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	A simply supported beam with overhang at one end is loaded as shown in the below Figure. Analyse and draw the Shear Force and Bending Moment Diagrams. Locate the points of contra-flexure. 	**5**	1	1,2,12	**4**
1B	A rectangular beam of cross-section 230 mm(wide) x 450 mm(deep) is simply supported over a span of 6 m. If the allowable bending stresses in the material is limited to 60 MPa, analyze maximum Point force at mid span that the beam can carry.	**3**	2	1,2,12	**4**
1C	A circular bar of length 300 mm and diameter 12 mm is subjected to twisting by a torque of 20 Kn-m. Analyse for shear stress in the bar	**2**	2	1,2,12	**3**
2A	 Using castiglianos strain energy Method analyse for Deflection under 20 kN point force. I = 10 x 10^{-6} m^4 and E = 200 GPa	**5**	4	1,2,12	**4**
2B	At a point in a strained material there is a tensile stress of 80 MPa upon the horizontal planes and compressive stress of 40 MPa upon the vertical planes. Shear stress of 48 MPa is acting	**3**	2	1,2,12	**4**

	on all these planes in positive direction. Determine the planes of maximum shear stress and magnitude of maximum shear stress.				
2C	A solid shaft is to transmit 337.5kW at 120 rpm. If the maximum shear stress is limited to 80MPa, calculate the diameter of the shaft	**2**	2	1,2,12	**3**
3A	Using unit-load method, analyse for horizontal displacement of the joint B of portal frame, ABC shown in the figure. 	**5**	5	1,2,12	**4**
3B	A symmetrical I section overall 250mm × 500mm, having 40mm thickness of flanges and web is used as a cantilever beam (with left end fixed) of span 3m. It carries a point load of 20kN at free end and an UDL of 5kN/m. Plot the bending stress distribution at 1m from the fixed end, given I = 1.3076 × 109 mm4.	**3**	2	1,2,12	**4**
3C	With example explain section modulus of a flexural member.	**2**	2	1,2,12	**2**
4A	A hollow column whose outside dimension 300mmx100mm has a thickness of 20mm. It is 4.0m long and is hinged at both ends. Calculate the safe load by Eular's and Rankine's formula using a factor of safety of 4. Take E=200GPa ,σc =500MN/m^2 , a=1/1650	**4**	3	1,2,12	**4**
4B	Determine the deflection under the point load on a simple supported beam of span 4m loaded with a downward point load 30kN at 3m from left support. Take EI = 9000kN-m2. Use Macaulay's double integration method.	**3**	4	1,2,12	**4**
4C	Determine maximum deflection at free end of cantilever beam 3m in length subjected to point force 25 KN at mid span using castiglianos theorem. EI = 12000kN-m^2	**3**	5	1,2,12	**4**

5A	Using conjugate beam method analyse for deflection and slope at free end of cantilever beam. assume I = 8 x 10^{-6} m^4 and E = 200 GPa 	**4**	4	1,2,12	**4**
5B	Analyse for vertical deflection at mid span of simply supported beam 6m length using unit load method subjected to UDL 20kN/m on left half of the span. 	**4**	5	1,2,12	**4**
5C	Define Maxwell's reciprocal theorem and Bettis Law used in unit load method.	**2**	5	1,2,12	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***5***
2	**15**
3	***4***
4	***12***
5	***14***

BL	Total Marks
2	**4**
3	**4**
4	**42**
5	